

## Performance Testing

|                      |                                   |
|----------------------|-----------------------------------|
| <b>Date</b>          | 15 March 2026                     |
| <b>Team ID</b>       | Group 5 (AP24110011158)           |
| <b>Project Name</b>  | Personalized Networking Assistant |
| <b>Maximum Marks</b> | 5 Marks                           |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                             |
|----------------------------|-----------------------------------------------------|
| <b>Testing Tool Used</b>   | [e.g., JMeter, Locust, k6, Postman]                 |
| <b>Type of Testing</b>     | [e.g., Load Testing, Stress Testing, Spike Testing] |
| <b>Target Module / API</b> | [Specify which feature or endpoint was tested]      |
| <b>Test Environment</b>    | [e.g., Local / Cloud / Hosted Server]               |
| <b>Test Date</b>           |                                                     |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome                |
|------|-----------------------------|----------------------|----------------|---------------------------------|
| 1    |                             |                      |                | Pytest benchmark timers         |
| 2    |                             |                      |                | Integration latency testing     |
| 3    |                             |                      |                | /generate-conversation endpoint |
| 4    |                             |                      |                |                                 |

### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                    |
|------|----------------------|--------------|--------------|----------------------|----------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 1.2 seconds  | Pass                 | Meets under 2s target      |
| 2    | Response Time (Max)  | < 5 seconds  | 1.8 seconds  | Pass                 | Meets under 5s target      |
| 3    | Throughput (Req/sec) | 5 req/sec    | 5 req/sec    | Pass                 | Offline CPU pipeline rate  |
| 4    | Error Rate           | < 1%         | 0% error     | Pass                 | 0 errors across 100 runs   |
| 5    | CPU Utilization      | < 80%        | 70% load     | Pass                 | Optimized CPU execution    |
| 6    | Memory Utilization   | < 80%        | 1.2GB RAM    | Pass                 | Transformers weights cache |

### Step 4: Observations & Analysis

**Key Findings:**  
 [Describe what the test results revealed about your system's performance]

**Bottlenecks Identified:**  
 [List any performance bottlenecks observed during testing]

**Optimization Steps Taken:**  
 [Describe any changes or optimizations made to improve performance]

### Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]